$f : \mathbb{R}^n \longrightarrow \mathbb{R}$

$F : \mathbb{R}^n \longrightarrow \mathbb{R}^n$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x) \ldots\ldots \frac{\partial f}{\partial x_n}(x) \right)$$ gradient

$$\nabla \cdot F(x) = \frac{\partial F_1}{\partial x_1}(x) + \ldots + \frac{\partial F_n}{\partial x_n}(x)$$ divergence

$$\nabla \times F(x) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} , \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} , \frac{F_2}{\partial x} - \frac{F_1}{\partial y} \right)$$

$n = 3$

$$\Delta f = \frac{\partial^2 f}{\partial x_1{}^2} + \ldots + \frac{\partial^2 f}{\partial x_n^2}$$

$n=3$

$\nabla \times \nabla f = 0$

$$\nabla \times \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2}{\partial z \partial y}, \ldots, \ldots\right) = \vec{0}$$

$\nabla \cdot (\nabla \times F) = 0$

$\nabla \cdot (-Y_z + Z_y, Z_x + X_z, -X_y + Y_x) =$

$Z_{yx} - Y_{zx} + X_{zy} - Z_{xy} + Y_{xz} - X_{yz} = 0$

cohomological complex, ke imaš dve zaporedni puščici je 0

$$\mathbb{R} \xrightarrow{\nabla} \mathbb{R}^3 \xrightarrow{\nabla \times} \mathbb{R}^3 \xrightarrow{\nabla \cdot} \mathbb{R}$$

$$K = [0,1] \times [0,1] \subseteq \mathbb{R}^2$$

$$\int_K \nabla \cdot \vec{F}\, dx\, dy = \int_{\partial K} \vec{F}\, d\vec{n}$$

normale

$$\int_0^1 \int_0^1 \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dx\, dy = \int_0^1 dy \int_0^1 \frac{\partial F_1(x,y)}{\partial x}\, dx + \int_0^1 dx \int_0^1 \frac{\partial F_2(x,y)}{\partial y}\, dy$$

$$= \int_0^1 F_1(x,y)\Big|_0^1 dy + \int_0^1 F_2(x,y)\Big|_0^1 dx =$$

$$= \int_0^1 \big(F_1(1,y) - F(0,y)\big)\, dy + \int_0^1 \big(F_2(x,1) - F_2(x,0)\big)\, dx$$

$$\int_{\partial K} \vec{F}\, d\vec{n} = \int_0^1 \big(F_1(x,0), F_2(x,0)\big) \begin{pmatrix} \overset{dx}{0}, & \overset{dy}{-1} \end{pmatrix} + \dots$$

$$= \int_0^1 -F_2(x,0)\, d\cancel{x} + \dots = \int_0^1 \big(F_1(1,y) - F_1(0,y)\, d\cancel{y} + \int_0^1 \big(F_2(x,1) - F_2(x,0)\, d\cancel{x}$$

$u_{xy} = x^2 y \qquad u(x,0) = x \qquad u(0,y) = y^2$

$u_x = \frac{1}{2}x^2y^2 + C(x)$

$u = \frac{1}{6}x^3y^2 + C(x) + D(y)$

$u(x,0) = C(x) + D(0) = x$

$u(0,y) = C(0) + D(y) = y^2$

$C(x) + D(y) = x + y^2$

$C(0)$ in $D(0)$ lahko damo za 0

$u(x,y) = \frac{1}{6}x^3y^2 + x + y^2$

$U_{xy} + U_x = y e^{-y}$ $\qquad u(0,y) = 0$

$u(x,x) = 0$

$u_y(0,y) = 0$

$\frac{u(x,x)}{\partial y} = 0$

$\rightsquigarrow$

integriramo po x

$U_y + U = xye^{-y} + C(y)$

v $(0,y) \rightsquigarrow$

$0 + 0 = 0 + C(y) = 0$

$U_y + u = xye^{-y}$

linearna diferencialna enačba

homogen del

$u + u_y = 0$

$u = -\frac{du}{dy}$ $\quad dy = -\frac{du}{u} \rightsquigarrow y = -\ln u + C(x)$

$u = Ce^{-y}$

partikularna

$C'(y)e^{-y} = xye^{-y}$

$C'(y) = xy$

$C(x,y) = \frac{1}{2}xy^2$

$u = \frac{1}{2}xy^2e^{-y} + C(x)e^{-y}$

$u(0,y) = C(0)e^{-y} = 0$

$u(x,x) = \frac{1}{2}x^3e^{-x} + C(x)e^{-x} = 0$

$\Rightarrow C(x) = -\frac{1}{2}x^3$

$u(x,y) = \frac{1}{2}x e^{-x}(y^2 - x^2)$

c)

$u_x + u_y = 0 \qquad u(x,0) = x^2$

$u(x,y) = \sum_{i,j} a_{ij} x^i y^j$

$u(x,0) = \sum_i a_{i0} x^i = x^2 \Rightarrow a_{20} = 1$

$a_{i0} = 0$ za $i \neq 2$

$u_x = \sum_{i,j} a_{ij}\, i\, x^{i-1} y^j = a_{10} + \sum_{i,j\geq 0} a_{i+1\,j} (i+1) x^i y^j$

$u_y = \sum_{i,j} a_{ij}\, j\, x^i y^{j-1} = a_{01} + \sum_{i,j\geq 0} a_{ij+1} (j+1) y^j x^i$

$u_x + u_y = a_{01} + a_{10} + \sum_{i,j\geq 1} (a_{i+1,j}(i+1) + a_{ij+1}(j+1)) x^i y^j = 0$

$\Rightarrow (a_{i+1})(i+1) + a_{ij+1}(j+1) = 0$ za $\forall\, xy$

$a_{01} + a_{10} = 0 \Rightarrow a_{01} = -a_{10}$

$\Rightarrow a_{0j} = 0$ razen $a_{02} = -2$

$a_{2,1} = a_{1,2}$

$a_{11} = -a_{0,2}$

$a_{11} = -a_{2,0}$

$a_{1,0} + a_{01} = 0,\ a_{1,1} + a_{02} = 0 \Rightarrow a_{11} = 2$

$j \backslash i$	0	1	2	3		
0	a_{00}	0	1	0	0	...
1	0	-2	0	0		
2	1	0	0			
3	0	0				
	0					

$a_{3,0} = -a_{2,1}$

$a_{11} = -a_{02}$

$u(x,y) = a_{00} + a_{2,0}' x^2 + a_{11} xy + a_{02} y^2$

$= a_{00} + x^2 - xy + y^2$

$u(x,0) = a_{00} + x^2 \Rightarrow a_{00} = 0$

$\Rightarrow u(x,y) = x^2 - xy + y^2$

$a_{2,0} = a_{11}$

Hmmm neki je narobe

$u(x,y) = x^2 - 2xy + y^2$

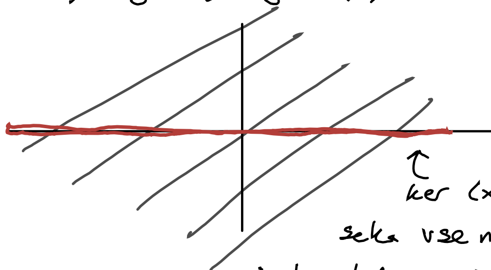
Pozabila sem množiti z $i+1$ in $j+1$ pri odvajanju

3. c)

$u_x + u_y = 0 \quad u(x,0) = x^2 \quad \rightsquigarrow u(x,y) = (x-y)^2$

Naj bo $u(x,y) = g(x-y)$

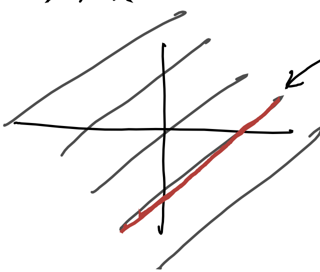
$\Rightarrow u_x + u_y = g'(x-y) - g'(x-y) = 0$



ker $(x,0)$ seka vse možnosti, je to validn način

Slab pimer

$u(x,x) = x^2$



če imamo tak robni pogoj ne moremo dobiti rešitve

Ali ne obstaja, ali pa jih je veliko

$$u_{xy} + y u_y + u = 1 \qquad u_x(x,0) = 1$$
$$u(x,x) = e^{-x^2}$$

$$y u_y + u = (yu)_y$$

$$u_{xy} + (yu)_y = 1 \rightsquigarrow u_x + yu = y + C(x)$$

$$V(x,0): \underbrace{u_x(x,0)}_{=0} + 0 \cdot 1 = 0 + C(x) \Rightarrow C(x) = 0$$

$$u_x + yu = y$$

homogeno:

$$u_x = -yu$$
$$\frac{\partial u}{u} = -y\,dx$$
$$u = D(y) e^{-yx}$$

partikularna:

$$D_x(x,y) e^{-yx} = y$$
$$\rightsquigarrow$$
$$D(x,y) = \int y e^{+yx}\,dx =$$
$$= e^{+yx} + C(y)$$

$$u = 1 + C(y) e^{-yx}$$
$$u(x,x) = 1 + C(x) e^{-x^2} = e^{-x^2} \Rightarrow C(x) = \frac{e^{-x^2} - 1}{e^{-x^2}} = 1 - e^{x^2}$$

$$u(x,y) = 1 + (1 - e^{-y^2}) e^{-yx}$$

$$u_{xy} + 2u_y = e^{-2x} \qquad u(x,0) = e^{-x} \qquad u(0,y) = 1$$
$$u_x(x,0) = -e^{-x}$$

---

integriramo po y

$$\rightsquigarrow \quad u_x + 2u = ye^{-2x} + C(x)$$

$$v\ (x,0): \quad -e^{-x} + 2e^{-x} = C(x) = e^{-x}$$

$$u_x + 2u = ye^{-2x} + e^{-x}$$

hom:

$$\frac{du}{u} = -2dx \Rightarrow u = e^{-2x} C(y)$$

part: $\quad C_x(x,y)e^{-2x} = ye^{-2x} + e^{-x}$

$$C(x,y) = \int (y + e^{x})dx = yx + e^{x} + D(y)$$

$$u = yxe^{-2x} + e^{-x} + e^{-2x}D(y)$$

$$v\ (0,y): \quad 0 + 1 + D(y) = 1 \Rightarrow D(y) = 0$$

$$\Rightarrow u = yxe^{-2x} + e^{-x}$$

KLE

$$a(x,y,u)u_x + b(x,y,u)u_y = c(x,y,u)$$

$S = \{ (x,y,z) \in \mathbb{R}^3 ; \quad z = u(x,y) \quad u \text{ rešitev} \}$

je integralska krivulja

Presek $S_1 \cap S_2$ je karakteristična krivulja

Karakteristična krivulja γ: $\dot{\gamma}(t) = (a(\gamma(t)), b(\gamma(t)), c(\gamma(t)))$

Izrek:

- S_1, S_2 integralski ploskvi za isto KLE. $p \in S_1 \cap S_2 \Rightarrow$ karakteristična krivulja, ki vsebuje p leži cela v $S_1 \cap S_2$
- $S_1 \cap S_2$, če se ne sekata tangentno, je karakteristična krivulja

Dokaz 2)

$p \in C = S_1 \cap S_2$

$\Pi_1 = T_p S_1 \qquad \Pi_2 = T_p S_2$

$\dim(\Pi_1 \cap \Pi_2) = 1$, ker se ne sekata S_1 in S_2 tangentno.

$\Pi_1 \cap \Pi_2$ tangentna na C

$$a(p), b(p), c(p) \in \Pi_1 \cap \Pi_2 = T_p C$$

$T_p C = \{\lambda (a(p), b(p), c(p)) + p \ ; \lambda \in \mathbb{R}\}$

C je tangentna na karakteristično smer

$\Rightarrow C$ je karak. krivulja

1

$$x u_x + y u_y = 0 \qquad (x, y, 0) = (a, b, c)$$

$$\dot{x} = x \rightsquigarrow x_0(s) e^t$$

$$\dot{y} = y \rightsquigarrow y_0(s) e^t$$

$$\dot{z} = 0 \rightsquigarrow z = z_0(s)$$

$$\varphi(t) = u(tx, ty)$$

$$\dot{\varphi}(t) = u_x \cdot x + u_y \cdot y = 0$$

$$\Rightarrow \varphi(t) = C(s) = C \qquad \varphi(0) = u(0,0)$$

$$\Rightarrow \varphi(t) = u(0,0)$$

$$\varphi(1) = u(x,y) = C = u(0,0)$$

$$\Rightarrow u(x,y) = u(0,0)$$

(2) $u : \bar{D} \to \mathbb{R}$

$a(x,y)\, u_x + b(x,y)\, u_y = -u$

$x a(x,y) + y b(x,y) > 0$ za $x,y \in \partial D$

min, max $D \Rightarrow u \equiv 0$

max $\in D \Rightarrow u_x = u_y = 0$ v (x_0, y_0)

$\Rightarrow -u(x_0,y_0) = 0$

min $\in D \Rightarrow -u(x_1, y_1) = 0 \Rightarrow$ $\Rightarrow u \equiv 0$

b) $(x,y,\lambda) \longmapsto u(x,y) + \lambda(x^2+y^2-1)$

$u_x(x,y) + 2\lambda x = 0$

$u_y + 2\lambda y = 0$

$x^2+y^2-1=0$

c) $t \overset{f}{\longmapsto} u(tx_0, ty_0)$ x_0, y_0 maks ~~...~~ na robu

$\dot{f} = u_x x_0 + u_y y_0 \geq 0$ v okolici $t=1$

$t=1: u_x(x_0,y_0)\, x_0 + u_y(x_0,y_0)\, y_0 = -2\lambda\, (x_0^2 + y_0^2)$

$\Rightarrow \lambda \leq 0$

Denimo da $u(x_0,y_0) > 0$

$a(x,y)\, (-2\lambda x_0) + b(x,y)\, (-2\lambda y_0) \overset{=-u}{<} 0$

$(-2\lambda)\, \underbrace{(a(x,y) x_0 + b(x,y) y_0}_{>0} < 0$

$\Rightarrow -2\lambda < 0 \Rightarrow \lambda > 0$ ⚡

Sledi $u(x_0,y_0) < 0$

ker kdi je minimum na robu $\geq 0 \Rightarrow$

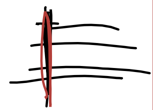
$u \equiv 0$

Metode karakteristik za enačbe 1. reda

enačba je

- $a(x, u(x)) \cdot Du = b(x, u(x))$
- $u(x) = g(x)$ .... začetni pogoj na krivulji Γ $\quad g : \Gamma \to \mathbb{R}$

① Rešimo sistem za vse y po s



odvod po s

$$\dot{z}(y,s) = b(x(y,s), z(y,s))$$

$$\dot{x}(y,s) = a(x(y,s), z(y,s))$$

$$x(y,0) = y \quad ; \quad z(y,0) = g(y)$$

② Najdemo $s(x)$, $y(x)$, potem $u(x) = z(y(x), s(x))$

(1) $u(0,x) = g(x)$

$u_t + a \cdot Du = 0$

$\dot{z}(y,s) = 0 \quad \rightsquigarrow \quad z = c(y)$

$\dot{t}(y,s) = 1 \quad \rightsquigarrow \quad t = 1s + c(y)$

$\dot{x}(y,s) = a \quad \rightsquigarrow \quad x = as + c(y)$

$(t(\cancel{y},0), x(y,0)) = (0,y)$

$z(y,0) = g(y)$

$z = c(y) = g(y)$

$t(y,0) = c(y) = 0 \rightsquigarrow t(y,s) = s$

$x(y,0) = c(y) = y \rightsquigarrow x(y,s) = as + y$

$y = x - at$

$u(x,y) = g(x-at)$

b)

$$u_t + a\nabla u + u = f(x)$$

$$u_t + a\nabla u = f(x) - u$$

$$\dot{z}(y,s) = f(x(y,s)) - z(y,s)$$

$$\dot{t}(y,s) = 1 \rightsquigarrow t = s + C(y)$$

$$\dot{x}(y,s) = a \rightsquigarrow x = s + D(y)$$

$$x(y,0) = g(y) \rightsquigarrow x = s + g(y)$$

$$t(y,0) = 0 \rightsquigarrow t = s$$

$$\rightsquigarrow \dot{z}(y,s) = f(s + g(y)) - z$$

hom: $\int \frac{dz}{z} = -\int ds \rightsquigarrow z = C(y) e^{-s}$

par: $\dot{C}(y,s) e^{-s} = f(s + g(y))$

$$\rightsquigarrow C = \int f(s + g(y)) e^{s}) ds$$

$$z(y,s) = \int f(t + g(y)) e^{t} dt \; e^{-s}$$

$$= \int f(t) e^{t} dt \; e^{-g(y)} e^{-s}$$

$$u(x,y) = \left(\int f(t) e^{t} dt\right) e^{-x}$$

c) $u_t + \nabla F(u) = 0$

$$\nabla F(u(x_1, \dots x_n)) = \left(F'(u) \frac{\partial u}{\partial x_1}, \dots, F'(u) \frac{\partial u}{\partial x_n} \right) =$$
$$= F'(u) \cdot \nabla u$$

$\dot{z} = 0 \quad \rightsquigarrow \quad z = C(y) = g(y)$

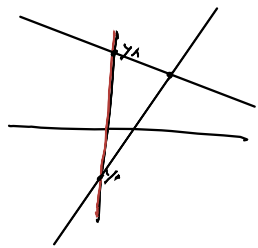
$\dot{x} = F'(z) \rightsquigarrow \dot{x} = F'(C) \rightsquigarrow x = F'(g(y))s + y$

$\dot{t} = 1 \quad \rightsquigarrow t = s + D(y) \rightsquigarrow t = s$

$t(y,0) = 0$

$x(y,0) = y$

$z(y,0) = g(y)$



$x(y,s) = F'(g(y))s + y$

Kaj ko se sekata

$$F'(g(y_0))s + y_0 = F'(g(y_1))s + y_0$$

če se sekata mora biti $g(\cancel{y_0}) = g(y_1)$

torej če se sekata, se ne more sekati
bi lahko
[illegible] rešitev [illegible]

a) $uu_x + u_y = 1 \qquad u(x,x) = \frac{x}{2}$

$a = u \qquad \dot{x} = z \rightsquigarrow x = \frac{1}{2}t^2 + \frac{s}{2}t + C(s) \rightsquigarrow$

$b = 1 \qquad \dot{y} = 1 \rightsquigarrow t + C(s) \rightsquigarrow y = t + s$

$c = 1 \qquad \dot{z} = 1 \rightsquigarrow t + C(s) \rightsquigarrow z = t + \frac{s}{2}$

$x(s,0) = s$

$y(s,0) = s$

$z(s,0) = \frac{s}{2}$

$x = \frac{1}{2}t^2 + \frac{1}{2}st + s \rightsquigarrow s = -$

$t = y - s \rightsquigarrow s = y - t$

$u(x,y) = \frac{s}{2} + t = \frac{s}{2} + y - s = y - \frac{s}{2}$

$x = \frac{1}{2}t^2 + \frac{1}{2}yt - t^2 + y - t = -\frac{1}{2}t^2 + \frac{1}{2}yt + y - t$

$\vdots \qquad s = \frac{3x - y^2}{2 - y}$

če $x \neq 2$ je vredu

oz. $\cancel{y} \neq 2$

b) $-\sin(y)\,u_x + x u_y = 0 \qquad u(x,0) = g(x) \quad$ za $x>0$

$$\left.\begin{aligned} \dot{x} &= -\sin(y) \\ \dot{y} &= x \end{aligned}\right\} \qquad \ddot{y} = \dot{x} = -\sin(y)$$

$$\frac{\partial^2}{\partial t^2}\sin(y) = \frac{\partial}{\partial t}\cos y \cdot \dot{y} =$$

$$\dot{z} = 0 \rightsquigarrow z = g(s) \qquad = \ddot{y}\cos y + \dot{y}(-\sin y)\dot{y}$$

$$x(s,0) = x$$

$$y(s,0) = 0$$

$$z(s,0) = g(s)$$

1\. energija $\quad E_1(x,y,z) = z$

$$\ddot{y} = \dot{x} = -\sin(y) \qquad x\dot{x} = \cancel{x}\ddot{y} = \dot{y}\ddot{y} = -x\sin(y) = -\dot{y}\sin y$$

2\. energija

$$x\dot{x} = (\cos y)^{\cdot}$$

$$\int x\,dx \int (\cos y)^{\cdot}\,dt = \cos y + C$$

$$\frac{1}{2}x^2 = \cos y + c$$

$$E_2 = \frac{1}{2}x^2 - \cos y$$

Iščemo H.

$$E_1(x(0), y(0), z(0)) = g(s)$$

$$E_2(x(s,0), y(s,0), z(s,0)) = \frac{1}{2}s^2 - 1$$

$$H(a,b) = a - g(\sqrt{2b+2})$$

$$H(E_1, E_2) = g(s) - g(\sqrt{s^2}) = 0$$

$$H(\underbrace{E_1(x,y,z)}_{=z}, E_2(x,y,z)) = 0 \quad \text{za vse } t,s$$

$$z - g\left(\sqrt{x^2 - 2\cos y + 2}\right) = 0$$

$$\Rightarrow u(x,y) = g\left(\sqrt{x^2 - 2\cos y + 2}\right)$$

c) $x(y^2+u)u_x - y(x^2+u)u_y = (x^2-y^2)u$

$u(x,x) = C \in \mathbb{R}$ za $x \neq 0$

$\dot{x} = x(y^2+z)$

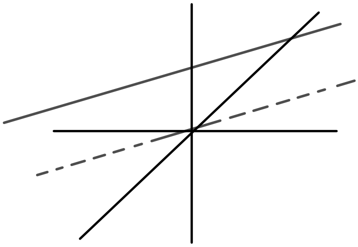
$\dot{y} = -y(x^2+z)$

$\dot{z} = z(x^2-y^2) = z(x-y)(x+y)$

$x(s,0) = s$

$y(s,0) = s$

$z(s,0) = c$



$\dot{x}(s,0) = s(s^2+c)$

$\dot{x}x + \dot{y}y = z(x^2-y^2) = \dot{z}$

$\frac{x^2}{2} + \frac{y^2}{2} = z + D(s)$

$E_1(x,y,z) = \frac{x^2}{2} + \frac{y^2}{2} - z$

$\frac{\dot{x}}{x} + \frac{\dot{y}}{y} + \frac{\dot{z}}{z} = \quad y^2 - x^2 + x^2 - y^2 = 0$

$\ln x + \ln y + \ln z = c$

$xyz = F(s) \qquad E_2(x,y,z) = xyz$

$E_1(s,0) = s^2 - c \quad = a$

$E_2(s,0) = cs^2 \quad = b \qquad ca - b + c^2$

$H(a,b) = ca - b + c^2$

$H(E_1,E_2) =$

$c\left(\frac{x^2}{2} + \frac{y^2}{2} - z\right) - xyz + c^2 = 0$

izrazimo z

$$z = u(x,y) = \frac{c^2 + \frac{c}{2}(x^2+y^2)}{c + xy}$$

Energija

Pri uporabi karakteristik v n-dimenzionalnem sistemu, lahko iščemo n "energij". To so funkcije $E_j(x(t))$ za katere velja, da so ~~konst~~

$$\frac{d}{dt} E_j(x(t)) = 0$$

Nato predpišemo $H(E_1, \ldots, E_n) = 0$ (oz poljubno konstanto) in z uporabo začetnih pogojev določimo H, ki zadošča $H(E_1(X(0)), \ldots) = 0$

12.3

$u_x + x u_y = u \qquad u(1,y) = h(y) \qquad h \in \mathcal{C}(\mathbb{R})$

$\dot{x} = 1 \qquad x(s,0) = 1$

$\dot{y} = x \qquad y(s,0) = \cancel{\ } s$

$\dot{z} = z \qquad z(s,0) = h(y)$

$x = t + 1$

$y = \frac{1}{2}t^2 + t + s$

$z = h(y)\, e^{x-1}$

$\dot{z} = z$

$\frac{dz}{z} = dt$

$z = ce^t = h(y) e^{x-1}$

$t = x - 1$

$s = y - \frac{1}{2}(x-1)^2 - (x-1)$

$u(x,y) = h\left(y - \frac{1}{2}(x-1)^2 - (x-1)\right) e^{x-1}$

$$x^2 u_x + y^2 u_y = (x+y)u \qquad u(x,1) = 1$$
$$x(s,0) = s$$

$$\dot{x} = x^2 \rightsquigarrow -\frac{1}{x} = t + c \rightsquigarrow x = \frac{-1}{t+c} \rightsquigarrow x = \frac{s}{1-ts}$$

$$\dot{y} = y^2 \rightsquigarrow y = \dots = \frac{1}{1-t}$$

$$\dot{z} = (x+y)z$$

$$\frac{\dot{z}}{z} = \frac{1}{1-t} + \frac{s}{1-ts}$$

nekje pride se zaseki, poglej

$$\ln z = -\ln(1-t) - \ln(1-ts) +$$

$$z = \cancel{\frac{1-ts}{1-t}} \frac{1}{(1-t)(1-ts)}$$

$$z = y \cdot \frac{x}{s}$$

$$s = x(1-ts)$$

$$s(1+tx) = x \rightsquigarrow s = \frac{x}{1-tx}$$

~~tz~~ $\cancel{(1-t)}$

$$(1-t)y = 1$$

$$t = -\frac{1}{y} + 1 = 1 - \frac{1}{y} = \frac{y-1}{y}$$

$$s = \frac{x}{1 - \frac{(y-1)x}{y}} = \frac{xy}{y+x-xy}$$

$$z = y + x - xy \rightsquigarrow u(x,y) = y + x - xy$$

$u(x,0)=1$

$$x(s,0)=s$$
$$y(s,0)=0$$
$$z(s,0)=1$$

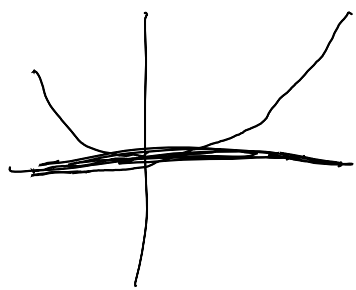
$$x=\frac{1}{t+c}$$
$$y=\frac{1}{t+c}$$

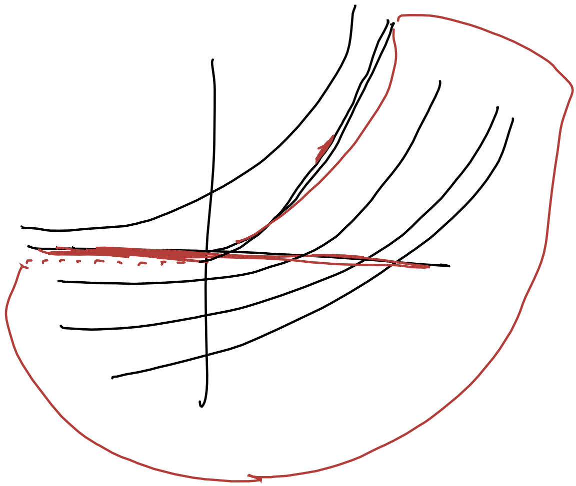
$$\frac{1}{c}=s \rightsquigarrow c=\frac{1}{s} \rightsquigarrow x=\frac{s}{1-ts}$$

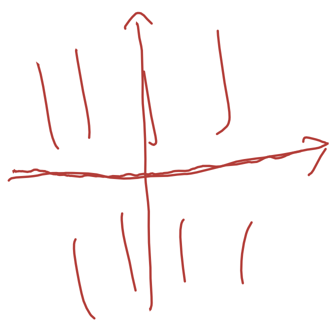
$$\frac{1}{c}=0 \rightsquigarrow c=\infty \qquad y=0$$

$$\dot{z}=\frac{s}{1-ts}z$$
$$\ln z=-\ln(1-ts)+C \rightsquigarrow$$

$$z=\frac{E(s)}{1-ts}=\frac{1}{1-ts}$$ ne moremo izraziti z x in y $\rightsquigarrow$ ne dobimo rešitve







ogrinjača družine ravnin skozi točko

Naj bo $\Pi_\lambda = \{(x,y,z) \mid z = \psi(x,y,\lambda) = a(\lambda)x + b(\lambda)y$
družina ravnin skozi izhodišče
$a,b \in \mathcal{C}(I)$ $I \subseteq \mathbb{R}$ interval, da je
$\vec{n}(\lambda) = (a(\lambda), b(\lambda), -1)$ injektivna
in $w(a',b') = \begin{vmatrix} a' & b' \\ a'' & b'' \end{vmatrix} \neq 0$

Opazimo: $z = -1$ komponenta je normalizacija, ki da $\Pi_{\lambda_1} = \Pi_{\lambda_2} \Leftrightarrow \lambda_1 = \lambda_2$

Cilj: Najdemo $\varphi(x,y)$, ki definira $\sigma = \{(x,y,\varphi(x,y))\}$ da velja
- $\sigma \subseteq \bigcup_\lambda \Pi_\lambda$
- $P_0 = (x_0, y_0, z_0) \in \sigma \cap \Pi_\lambda$ za nek λ

$\Longrightarrow$ normalni vektor za σ v točki p_0 je sorazmeren z $\vec{n}(\lambda)$

Opazka: $\sigma = \Pi_\lambda$ $\overset{(\bigcup \Pi_\mu)}{\text{za}}$ fiksen λ je trivialna rešitev, ki jo ignoriramo

ker je $\vec{n}(\lambda)$ injektivna, če $p_0 \in \sigma \cap \Pi_\lambda$ in $p_0 \in \sigma \cap \Pi_\mu$ potem $\lambda = \mu$

Consider the map $P: \mathbb{R}^3 \to \mathbb{R}^2$
če $(x,y) = P(x,y,z)$ za nek (x,y,z) $P \in \sigma$
potem $p \in \Pi_\lambda$ za nek unikaten $\lambda \subset \Lambda(x,y)$

$$\Lambda : P(\sigma) \to \mathbb{R}$$
$$(x,y) \mapsto \Lambda(x,y)$$

Ker $\sigma = \{z = \varphi(x,y)\} \subseteq \bigcup_\lambda \Pi_\lambda$ imamo
$$\varphi(x,y) = a(\Lambda(x,y))x + b(\Lambda(x,y))y = \psi(x,y,\Lambda(x,y))$$

normala za σ je paralelna $\vec{n}(\lambda)$, torej
parametrizacija σ je $u(x,y) = (x,y,\varphi(x,y))$
normala je podana z $u_x \times u_y$
računamo:
$$u_x = (1, 0, \varphi_x(x,y) = \psi_x(x,y,(\Lambda(x,y)) + \psi_\lambda(x,y,(\Lambda(x,y))\Lambda_x(x,y))$$
$$u_y = (0, 1, \varphi_y(x,y) = \psi_y(x,y,\Lambda(x,y)) + \psi_\lambda(x,y,\Lambda(x,y))\Lambda_y(x,y))$$

$$-u_x \times u_y = (\psi_x + \psi_\lambda \Lambda_x, \psi_y + \psi_\lambda \Lambda_y, -1)$$
$$= (\psi_x, \psi_y, -1) + \psi_\lambda \cdot (\Lambda_x, \Lambda_y, 0) =$$

Spomnimo se $\psi = a(\lambda)x + b(\lambda)y$

$$= \underbrace{a(\Lambda(x,y)), b(\Lambda(x,y)), -1)}_{\vec{n}(\Lambda(x,y))} + \psi_\lambda'(\Lambda_x, \Lambda_y, 0)$$

Želimo da je $= 0$

1. $(\Lambda_x, \Lambda_y, 0) = 0 \Rightarrow D\Lambda = 0 \Rightarrow \Lambda$ konstantna
   $\Rightarrow \psi(x,y) = a(\Lambda)x + b(\lambda)y \Rightarrow \sigma \subseteq \Pi_\lambda$ trivialna rešitev

2. $\psi_\lambda(x,y,\Lambda(x,y)) = 0$ $\forall x,y \in P(\sigma)$

vseen [illegible]

a)

$(u_x u_y)^2 = u \qquad u(x,0) = x$

$(pq)^2 = z$

$F(x,y,z,p,q) = (pq)^2 - z$

$\dot{x} = \cancel{\ } = 2(pq)q$

$\dot{y} = 2(pq)p$

$\dot{z} = 2p^2q^2 + 2p^2q^2 = 4p^2q^2$

$\dot{p} = -(-p) = p$

$\dot{q} = q$

$p = C(s)e^t$

$q = De^t$

$z = 4\dot{z} \rightsquigarrow z = E(s)e^{4t}$

$\dot{z} = 4e^{4t}CD$

$z = e^{4t}C^2D^2 = (e^{4t}-1)C(s)^2D(s)^2 + E(s)$

$\dot{x} = 2C^2De^{3t} \rightsquigarrow x = \frac{2}{3}C^2D(e^{3t}-1) + F(s)$

$\dot{y} = \frac{2}{3}CD^2e^{3t} \rightsquigarrow x = \frac{2}{3}CD^2(e^{3t}-1) + G(s)$

$u(x,0) = x$

$x(s,0) = s = F(s)$

$y(s,0) = 0 = G(s)$

$z(s,0) = s = E(s)$

$p(s,0) = u_x(x(s,0), y(s,0)) = 1 \rightsquigarrow C(s) = 1$

$q(s,0) = \pm\sqrt{s}$

$(u_x u_y)^2(x,0) = u(x,0) = s$

$(1 \cdot u_y(x,0))^2 = x$

$q_0 = u_y(x,0) = \pm\sqrt{s}$

$u(x,0) = \pm\sqrt{x}\,y + c(x)$

$z = (e^{4t}-1)s + s$

$x = \frac{2}{3}(e^{3t}-1)s + s \qquad s = \frac{x}{1+\frac{2}{3}(e^{3t}-1)}$

$y = \frac{2}{3}(e^{3t}-1)\sqrt{s}$

$p = e^t \qquad e^{3t} = \frac{3}{2}\frac{y}{\sqrt{s}} + 1$

$q = e^t\sqrt{s}$

$s = \frac{x}{1+\frac{y}{\sqrt{s}}}$

$s + \sqrt{s}\,y = x$

$s = \left(\frac{y}{\frac{2}{3}(e^{3t}-1)}\right)^2$

$x = \frac{y^2}{\underbrace{\frac{2}{3}(e^{3t}-1)}_{v}} + \left(\frac{y}{\frac{2}{3}(e^{3t}-1)}\right)^2$

$s = \left(\frac{x^2}{1 \pm \sqrt{y^2-4x}}\right)^2$

$v^2x - vy^2 - y^2 = 0$

$D = \sqrt{y^4 - 4xy^2} = y\sqrt{y^2-4x}$

$\frac{2}{3}(e^{3t}-1) = \frac{y^2 \pm y\sqrt{y^2-4x}}{x^2}$

$u = \frac{3}{2}\left(\frac{y^2 \pm x\sqrt{y^2-4x}}{x^2}\right)\left(\frac{x^2}{1\pm\sqrt{y^2-4x}}\right)^2 + \left(\frac{x^2}{1\pm\sqrt{y^2-4x}}\right)^2$

b) $u_x^2 + u_y = yu$ $\qquad u(x,0) = x$

$F(x,y,z,p,q) = p^2 + q - yz$

$\dot{x} = 2p$ $\qquad x(s,0) = s$

$\dot{y} = 1$ $\qquad y(s,0) = 0$

$\dot{z} = p\cdot(2p) + q\cdot 1$ $\qquad z(s,0) = s$

$\dot{p} = -0 + py = py$ $\qquad p(s,0) = 1$

$\dot{q} = z + qy$ $\qquad q(s,0) = -1$

$y = t + c(s) = t$

$dp = pt\,dt \rightsquigarrow \ln p = \frac{1}{2}t^2 dt + c \rightsquigarrow p = p_0 e^{\frac{1}{2}t^2} = e^{\frac{1}{2}t^2}$

$dx = 2e^{\frac{1}{2}t^2} dt \rightsquigarrow x =$

$2\int e^{\frac{1}{2}t^2} dt = \int \sqrt{2}\, u^{-\frac{1}{2}} e^u \, du$

$u = \frac{1}{2}t^2$

$du = t\,dt = \sqrt{2u}\,dt$

$\dot{z} = 2e^{t^2} + q$

$\dot{q} = z + qt$

$\ddot{z} = \ddot{q} + q + t\dot{q} = 2e^{t^2} q$

$\ddot{q} + t\dot{q} = 2e^{t^2}$

hom: $\frac{d\dot{q}}{\dot{q}} = -t\,dt \rightsquigarrow \dot{q} = A(s)e^{-t}$

part: $A' e^{-t} = 2e^{t^2}$

$A' = 2e^{t^2+t}$

b) $u_x^2 + u_y = yu$ $\qquad u(x,0) = x$

$F(x,y,z,p,q) = p^2 + q - yz$

$$\dot{x} = 2p \qquad x(s,0) = s$$
$$\dot{y} = 1 \qquad y(s,0) = 0$$
$$\dot{z} = p\cdot(2p) + q\cdot 1 \qquad z(s,0) = s$$
$$\dot{p} = -0 + py = py \qquad p(s,0) = 1$$
$$\dot{q} = z + qy \qquad q(s,0) = -1$$

Linearne PDE 2. reda v $\mathbb{R}^2$

$$\underbrace{a(x,y)u_{xx}+2b\,u(x,y)u_{xy}+cu_{yy}}\ +du_x+eu_y+fu+g=0$$

$$\left(\frac{\partial}{\partial x},\frac{\partial}{\partial y}\right)\begin{pmatrix} a & b \\ b & c\end{pmatrix}\begin{pmatrix}\frac{\partial}{\partial x}\\ \frac{\partial}{\partial y}\end{pmatrix}u$$

$$\Delta = b^2-ac=-\det\begin{pmatrix} a & b\\ b & c\end{pmatrix}$$

kanonična oblika

$\Delta>0$ hiperbolični tip $\qquad u_{xy}+\text{nižji red}=0$

oziroma $\quad u_{xx}-u_{yy}+\text{niž.}=0$

$\Delta=0$ parabolični tip $\qquad u_{xx}+\text{niž}=0$

$\Delta<0$ eliptični tip $\qquad u_{xx}+u_{yy}+\text{niž.}=0$

Hiperbolični tip:

$$\frac{\partial y}{\partial x}=\frac{b\pm\sqrt{\Delta}}{a}\ \rightsquigarrow\ y=f_\pm(x,c)$$

Novi koordinati $\xi(x,y)$, $\eta(x,y)$

veljа $\ y=f_+(x;\xi(x,y))$, $y=f_-(x,\eta(x,y))$

Primer: $\frac{\partial y}{\partial x}=2\pm1 \rightsquigarrow y=\begin{cases}3x+c=f_+\\ x+c=f_-\end{cases}$

$\xi(x,y)=y-3x$

$\eta(x,y)=y-x$

- če $a=0 \rightsquigarrow \frac{dx}{dy}=\frac{b\pm\sqrt{D}}{c}$
- če $a=0$ in $c=0$ $\rightsquigarrow$ enačba je že v kanonični obliki

Parabolični tip $\Delta = 0$

$$\frac{dy}{dx} = \frac{b}{a} \quad \left(= \frac{b \pm \sqrt{\Delta}}{a}\right) \rightsquigarrow y = f(x, C)$$

$$y = f(x, \xi(x,y))$$

$\eta(x,y)$ je poljubna (ampak še koordinati)

Eliptični tip $\Delta < 0$

$$\frac{dy}{dx} = \frac{b \pm i\sqrt{-\Delta}}{a} = \frac{b \pm \sqrt{\Delta}}{a} \rightsquigarrow y = f(x, c)$$

Iščemo $\Phi(x,y)$: $y = f(x, \Phi(x,y))$

$$\xi = \operatorname{Re}\Phi \quad \eta = \operatorname{Im}\Phi$$

$$u_{xx} + 2u_{xz} + u_{yy} + u_{zz} = 0$$

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Diagonaliziramo

$$(\lambda I - A) = \begin{bmatrix} \lambda-1 & 0 & -1 \\ 0 & \lambda-1 & 0 \\ -1 & 0 & \lambda-1 \end{bmatrix} =$$

$$= (\lambda-1)\left((\lambda-1)^2 - 1\right) = (\lambda-1)(\lambda^2 - 2\lambda) =$$
$$\lambda(\lambda-1)(\lambda-2)$$

$\lambda = 0: \quad e_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ $\qquad Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$\lambda = 1: \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\lambda = 2: \quad e_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ $\qquad D = \begin{bmatrix} 0 & & \\ & 1 & \\ & & 2 \end{bmatrix}$

$$A = Q^T D Q$$

$\xi_1 = x - z$

$\xi_2 = y$ $\qquad \xi = Q \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$\xi_3 = x + z$

$$u = u\,(x(\xi), y(\xi), z(\xi)) = V(\xi)$$

$\Updownarrow$

$x = \frac{1}{2}(\xi_1 + \xi_3)$

$u_x = (\xi_1)_x V_{\xi_1} + (\xi_2)_x (V_{\xi_2}) + (\xi_3)_x V_{\xi_3}$ $\qquad y = \xi_2$

$z = \frac{1}{2}(\xi_3 - \xi_1)$

...

$u_x = V_{\xi_1} + V_{\xi_3}$ $\qquad \partial_x = \partial_{\xi_1} + \partial_{\xi_3}$

$u_y = V_{\xi_2}$ $\qquad \partial_y = \partial_{\xi_2}$

$u_z = -V_{\xi_1} + V_{\xi_3}$ $\qquad \partial_z = \partial_{\xi_3} - \partial_{\xi_1}$

$u_{xx} = V_{\xi_1} + 2V_{\xi_1\xi_3} + V_{\xi_3}$

$u_{yy} = V_{\xi_2\xi_2}$

$u_{zz} = V_{\xi_1} \cancel{-2V_{\xi_1\xi_3}} + V_{\xi_3\xi_3}$

$u_{xz} = -V_{\xi_1\xi_1} + \ldots$

$$u_{xx} + 2u_{xz} + u_{yy} + u_{zz} = 0$$

$\Rightarrow \quad 4V_{22} + V_{11} = 0$ $\qquad$ ni kanonična oblika

Morali bi normirati Q

$\tilde{\xi}_2 = \frac{\xi_2}{2} \quad \rightsquigarrow \quad \cancel{V_{22}} = \frac{\xi_2}{2}$ $\qquad \tilde{V}(\xi_1, \tilde{\xi}_2, \xi_3)$

$= V(\xi_1, \frac{\xi_2}{2}, \xi_3$

$\tilde{V}_2 = V_2 \cdot \frac{1}{2}$

$\tilde{V}_{22} = V_{22} \cdot \frac{1}{4}$

$$\Rightarrow \quad \tilde{V}_{22} + V_{11} = 0$$

D'Alzimbertova formula na $[0,\infty) \times \mathbb{R}$

$$u(t,x) = \frac{1}{2}\left(f(x-t) + f(x+t)\right) + \frac{1}{2}\int_{x-t}^{x+t} g(\xi)\,d\xi + \frac{1}{2}\iint_{\Delta} F(\tau,\xi)\,d\tau\,d\xi$$

kjer je $\Delta(t,x) = \{(\tau, \xi) \in \mathbb{R}^2 \mid 0 \le \tau \le t,\ |\xi - x| \le t-\tau\}$

$$u_{tt} - u_{xx} = F(t,x)$$
$$u(0,x) = f(x)$$
$$u_t(0,x) = g(x)$$

$$u_{tt} - u_{xx} = 0$$
$$u(0,x) = u_t(0,x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

$k, \omega > 0$

$$u_{tt} - u_{xx} = \sin(kx)\cos(\omega t)$$

$$u(0,x) = u_t(0,x) = 0$$

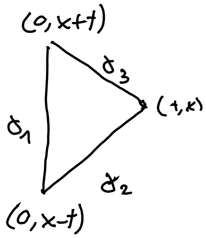
$$f(x) = g(x) = 0$$

$$F(x,t) = \sin(kx)\cos(\omega t)$$

$$u(t,x) = \frac{1}{2} \iint_{\Delta} F(\tau,\xi)\, d\tau\, d\xi$$

$$\Delta = \{ (\tau,\xi) \in \mathbb{R}^2 \mid 0 \le \tau \le t,\ |\xi - x| \le t - \tau \}$$

$$-\iint_D (Y_x - X_y)\, dx\, dy = \int_{\partial D} (X,Y)\, ds$$



$$X = -\frac{1}{\omega}\cos(kx)\cos(\omega t)$$

$$Y = 0$$

$$X_y = \sin(kx)\cos(\omega t)$$

$$\iint_{\Delta} \dots = -\int \left(-\frac{1}{\omega}\cos(kx)\cos(\omega t), 0\right) ds$$

$\gamma_1 : 0$

$\gamma_2 :\ \gamma_1(s) = (s, x-(t-s)) \quad s \in (0,t)$

$\gamma_3 = (t-s, x+s) \quad s \in (0,t)$

$$\int_{0}^{1} \left(\frac{1}{k}\cos(\omega s)\cos(k(x-t+s)), 0\right) \cdot (1,1)\, ds =$$

$$= \int \frac{1}{2}\left(\cos(\alpha+\beta) + \cos(\alpha-\beta)\right) =$$

$$= \int_0^1 \frac{1}{2}\left(\cos(s(\omega+k)+x-t) + \cos(s(\omega-k)-x+t)\right) ds$$

$$u_{tt} - u_{xx} = 0$$

$$u(0,x) = u_t(0,x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

d'Alembertova formula

za velike t

$$u(x,y) = 0 + \frac{1}{2}\int_{-1}^{1} 1 \, d\xi + 0 = 1$$

$$\lim_{t \to \infty} = 1$$

$$u(x,t) = \frac{f(x-t) + f(x+t)}{2}$$

$$\sup = ?$$

$$u(x,t) \leq \frac{1}{2}(1+1) + \frac{1}{2} \cdot 2 + 0 = 2$$

$$x + t = m < 1 \qquad x > 0 \Rightarrow t < 1$$

$$u(x,t) = \frac{1}{2}(1+1) + \frac{1}{2}\int_{x-t}^{x+t} g(\xi)\, d\xi$$

$$x = 0 \qquad t = m < 1$$

$$u(x,t) = 1 + \frac{1}{2}\int_{-m}^{m} g(\xi)\, d\xi = 1 + m$$

$$\lim_{m \to 1} u(x,t) = 1 \Longrightarrow \sup = 2$$

$$u_{tt} - u_{xx} = 0$$

$g(x) = 0$ $\quad t, x \in [0, \infty]$

$u(t, 0) = h(t)$

$u(t, 0) = h(t)$

$$u(x,t) = \frac{f(x-t) + f(x+t)}{2}$$

$$u(t,0) = \frac{1}{2}\left(f(t) + f(-t)\right) = h(t)$$

??

$u_{tt} - u_{xx} = 0$ $\quad$ $u(0,x) = f(x)$

$u_t(0,x) = 0$

$u(t,0) = h(t)$

$\delta = 0 + 1 = 1$

$$\frac{dx}{dt} = \frac{0 \pm \sqrt{1}}{1} = \pm 1$$

$x = \pm t + c$ $\quad$ $\xi(t,x) = x+t$

$\eta(t,x) = x-t$

$w_{\xi\eta} = 0$

$w_\xi = C(\xi)$

$w = C(\xi) + D(\eta)$

$u = \underbrace{C}_{F}(x+t) + \underbrace{D}_{G}(x-t)$

$x, t > 0$

$u(0,x) = F(x) + G(x) = f(x)$

$u(t,0) = F(t) + G(-t) = h(t)$

$u_t(0,x) = F'(x) - G'(x) = 0$

$F'(x) = G'(x) \rightsquigarrow F(x) = G(x) + C$

$$G(x) = \frac{f(x) - C}{2}$$

$$F(x) = \frac{f(x) + C}{2}$$

$$F(x) + G(x) = \frac{2f(x)}{2} \rightsquigarrow c \text{ je lahko } 0$$

za $x > 0$ vemo

$F(x) = \frac{1}{2} f(x)$

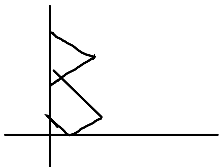
$G(x) = \frac{1}{2} f(x)$

$F(x) = h(x) - G(-x) = \frac{1}{2} f(x)$

$\rightsquigarrow G(x) = h(-x) - \frac{1}{2} f(-x)$ za $x < 0$

Splošna rešitev:

$$u(x,y) = \begin{cases} \frac{1}{2}(f(x+t) + f(x-t)) & ; \ x-t > 0 \\ \frac{1}{2} f(x+t) - \frac{1}{2} f(-x+t) + h(-x+t) & ; \ x-t < 0 \end{cases}$$



$$u_{tt} - u_{xx} = 0 \qquad t > 0 \quad x \in \mathbb{R}$$

$$u(0,x) = x$$

$$u_t(0,x) = 0$$

iščemo w, da bo $w_{tt} - w_{xx} = 0$

$$w = u + \frac{1}{12}x^4 - x \qquad w_t = u_t$$

$$w_x = u_x + \frac{1}{3}x^3 - 1 \qquad w_{tt} = u_{tt}$$

$$w_{xx} = u_{xx} + x^2$$

$$w(0,x) = u(0,x) + \frac{1}{2}x^4 - x = \frac{1}{2}x^4$$

$$w_t(0,x) = u_t(0,x) = 0$$

$$w = F(x+t) + G(x-t)$$

$$w(x,0) = F(x) + G(x) = \frac{1}{2}x^4$$

$$w_t(0,x) = F'(x) - G'(x) = 0$$

$$\rightsquigarrow F(x) = G(x) + C$$

$$2G(x) + C = \frac{1}{2}x^4$$

$$G(x) = \frac{1}{4}x^4 - \frac{1}{2}C$$

$$F(x) = \frac{1}{4}x^4 + \frac{1}{2}C$$

$$w(x,t) = \frac{1}{4}\left((x+t)^4 + (x-t)^4\right)$$

$$u(x,t) = \frac{1}{4}\left((x+t)^4 + (x-t)^4\right) - \frac{1}{12}x^4 + x$$

u, v resitvi $\quad u_{tt} - u_{xx} = 0 \quad t > 0$

$u_x(t,a) = u_x(t,b) = 0$

$$u_t v_t + u_x v_x \overset{?}{=} F(x+t) + G(x-t)$$

$u = f(x+t) + g(x-t)$

$v = h(x+t) + k(x-t)$

$$u_t v_t + u_x v_x = (f'(x+t) - g'(x-t))(h'(x+t) - k'(x-t))$$
$$+ (f'(x+t) + g'(x-t))(h'(x+t) + k'(x-t)) =$$
$$= \underbrace{2f'(x+t)h'(x+t)}_{F(x+t)} + \underbrace{2g'(x-t)k'(x-t)}_{G(x-t)}$$

$$F: t \mapsto \int_a^b u_t v_t + u_x v_x$$

$$F' \overset{!}{=} \int_a^b \frac{\partial}{\partial t} (u_{tt} v_t + u_t v_{tt} + u_{xt} v_x + u_x v_{xt}) \, dx =$$

$$= \int_a^b (u_{tt} v_t + u_t v_{tt}) \, dx + \underbrace{u_t v_x \Big|_a^b}_{0} + u_t v_x \Big|_a^b - \int_a^b (u_t v_{xx} + u_{xx} v_t) \, dx$$

$$= \int_a^b v_t (\underbrace{u_{tt} - u_{xx}}_{0}) + u_t (\underbrace{v_{tt} - v_{xx}}_{0}) \, dx = 0$$

$$\underbrace{u_x^2 u_{xx}}_{\left(\frac{1}{3}u_x^3\right)_x} \pm 2u_x u_y u_{xy} + \underbrace{u_y^2 u_{yy}}_{\left(\frac{1}{3}u_y^3\right)_y} = 0$$

$u = f(x)g(y)$

$u_x = f'(x)g(y)$ $\quad u_y = f(x)g'(y)$

$u_{xx} = f''(x)g(y)$ $\quad u_{yy} = f(x)g''(y)$ $\quad u_{xy} = f'(x)g'(y)$

$\rightsquigarrow f'^2(x)g^3(y)f''(x) \pm 2f'^2(x)g(y)f(x)g'^2(y) + g'^2(y)f^3(x)g''(y)$

delimo z $f^3 g^3$

$\rightsquigarrow$

$$\frac{f'^2 f''}{f^3} \pm \frac{2f'^2(x)\,g'^2(y)}{g^2(y)f^2(x)} + \frac{g'^2 g''(y)}{g^3(y)} =$$

Nastavek $f(x) = e^{\lambda x}$ $\rightsquigarrow \lambda e^{\lambda x} \rightsquigarrow \lambda^2 e^{\lambda x}$

$g(x) = e^{\mu y}$

$$\lambda^4 \pm 2\lambda^2\mu^2 + \mu^4 = 0$$

$$\rightsquigarrow (\lambda^2 \pm \mu^2)^2 = 0$$

$-$: $(\lambda-\mu)(\lambda+\mu)(\lambda^2+\mu^2) = 0 \rightsquigarrow \lambda = \pm\mu$

$+$: $\lambda^2 + \mu^2 = 0 \rightsquigarrow \lambda = 0 \wedge \mu = 0$

$u)$ $\quad u = e^{\lambda x} e^{\pm\lambda x} = e^{\lambda(x \pm y)}$

$u_{tt} - u_{xx} = 0 \qquad t\in(0,\infty)$

$u(0,x) = f(x) \qquad x\in(0,1)$

$u_t(0,x) = 0$

$u(t,0) = u(t,1) = 0$

$u(x,t) = F(x-t) + G(x+t)$

$u(0,x) = F(x) + G(x) = f(x)$

$u_t(0,x) = -F'(x) + G'(x) = 0$

$u(t,0) = F(-t) + G(t) = 0$

$u(t,1) = F(1-t) + G(1+t) = 0$

$F(x) = G(x) + c$

$F(x) = \frac{f(x)-c}{2}$

$G(x) = \frac{f(x)+c}{2} \rightsquigarrow c=0 \rightsquigarrow \quad F(x) = \frac{f(x)}{2}, \quad G(x) = \frac{f(x)}{2}$

To je za $x\in(0,1)$

$-F(-t) = G(t) \rightsquigarrow f(-t) = -f(t)$ za $t\in(0,1$

$\cancel{u(t,1)=}\ F(1-t) = -G(1+t)$

$t\in(0,1) \rightsquigarrow$

$f(1-t) = -G(1+t)$

$x = 1+t \rightsquigarrow -t = 1-x$

$G(t) = f(2-t)$ za $t\in(1,2)$

$t\in(n,n+1) \rightsquigarrow x\in(n,n+1)$

$G(1+t) = -F(1-t)$

$G(x) = -F(2-x)$

$$G(x) = \begin{cases} -F(2-x) & ; x \geq 1 \\ \frac{1}{2}f(x) & ; x\in(0,1) \\ -\frac{1}{2}f(-x) & ; x<0 \end{cases}$$

$t\in(0,2) \rightsquigarrow F(1-t) = -G(1+t)$

$s\in(1,3) \rightsquigarrow F(s) = -G(2-s) = -f(2-s)\cdot\frac{1}{2}$

$$F(x) = \begin{cases} -\frac{1}{2}f(-x) & ; x\in(-1,0) \\ \frac{1}{2}f(x) & ; x\in(0,1) \\ -\frac{1}{2}f(2-x) & ; x\in(1,2) \\ \frac{1}{2}f(2+x) & ; x\in(2,3) \end{cases}$$

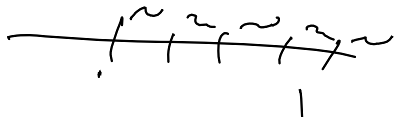
$s = 1-t$

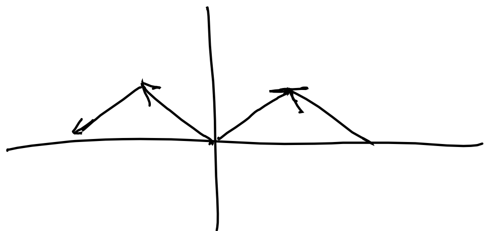
$t\in(-n-2,-n) \qquad F(1-t) = -G(1+t)$

$\rightsquigarrow s\in(n-1,n+1) \quad \underline{F(s)} = -G(2-s) \underline{=} F(s-2)$

$s\in(2n,2n+1) \rightsquigarrow F(s) = F(s-2) = F(s-2n) = \frac{1}{2}f(s-2n)$

$s\in(2n-1,2n) \rightsquigarrow F(s) = F(s-2) = F(s-2n) = -\frac{1}{2}f(2n-s)$





$$u_t - k u_{xx} = 0 \qquad t \in (0, \infty)$$
$$x \in [0, 2\pi)$$

$$u(t,0) = u(t,2\pi)$$
$$u_x(t,0) = u_x(t,2\pi)$$
$$u(0,x) = f(x)$$

$$u(x,t) = F(x - \sqrt{k}t) + G(x + \sqrt{k}t) \qquad c = \sqrt{k}$$

$$u(0,x) = F(x) + G(x) = f(x)$$
$$u_x = F' + G'$$
$$F'(-ct) + G'(ct) = F'(2\pi - ct) + G'(2\pi + ct)$$
$$-F(-ct) + G(ct) = -F(2\pi - ct) + G(2\pi + ct) + D$$

$u_t - u_{xx} = 0 \qquad (t,x) \in (0,\infty)\times I \qquad I \subseteq \mathbb{R}$

z robnimi in začetnimi pogoji:

Pišemo: $u(t,x) = v(t)\,w(x)$

$\Rightarrow v'(t)\,w(x) = v(t)\,w''(x)$

$\Rightarrow \frac{v'}{v}(t) = \frac{w''}{w}(x) \equiv \lambda \in \mathbb{R}$

$\Rightarrow v(t) = ce^{\lambda t}$

$w(x) = \alpha e^{\sqrt{\lambda}x} + \beta e^{-\sqrt{\lambda}x} \qquad \alpha,\beta \in \mathbb{C}$

enačba je linearna $\Rightarrow$

$u(t,x) = \sum_{k=0}^{\infty} v_k(t)\,w_k(x)$

Odločimo $W_k(x) = \alpha_k e^{\sqrt{\lambda_k}x} + \beta_k e^{-\sqrt{\lambda_k}x}$, $v_k = c_k e^{\lambda_k t}$

Določimo λ_k odvisno od robnega pogoja

in c_k odvisno od začetnega pogoja

$u(t,a) = u(t,b) = 0$

$V = \{ w \in L^2([a,b],\mathbb{R})\,;\, w(a) = w(b) = 0 \}$

$$u_t - k u_{xx} = 0 \qquad t \in (0,\infty), \quad x \in [0, 2\pi)$$

$$u(t,0) = u(t,2\pi)$$
$$u_x(t,0) = u_x(t,2\pi)$$
$$u(0,x) = f(x) = \sin x$$

$$u(t,x) = \sum v_k(t) w_k(x)$$

$$u(t,0) = u(t,2\pi) \Rightarrow \sum v_k(t) w_k(0) = \sum v_k(t) w_k(2\pi)$$
$$\Rightarrow \sum v_k(t)\left(w_k(0) - w_k(2\pi)\right) = 0$$
$$u_x(t,x) = \sum v_k(t) w_k'(x) \Rightarrow$$
$$\sum v_k(t)\left(w_k'(0) - w_k'(2\pi)\right) = 0$$
$$u(0,x) = f(x) \Rightarrow \sum v_k(0) w_k(x) = f(x)$$

$$v_k = c_k e^{\lambda_k t} \rightsquigarrow \sum c_k e^{\lambda_k t}\left(w_k(0) - w_k(2\pi)\right) = 0$$

$$w_k(0) = \alpha_k + \beta_k = 0$$
$$w_k(\pi) = \alpha_k e^{\sqrt{\lambda}\pi} + \beta_k e^{-\sqrt{\lambda}\pi} = 0$$
$$e^{\sqrt{\lambda}\pi} - e^{-\sqrt{\lambda}\pi} = 0$$

$$e^{2\sqrt{\lambda_k}\pi} = 1 \Rightarrow 2\sqrt{\lambda_k}\pi = 2\pi i k$$
$$\sqrt{\lambda_k} = ik$$
$$\lambda_k = -k^2$$

Pomanjkanje samo razmerje med α in β

$$w_k = e^{ik\cancel{\pi}x} - e^{-ik\cancel{\pi}x} \rightsquigarrow \cancel{\text{BSES}} \rightsquigarrow w_k = \alpha_k i \sin kx$$

$$\underline{w_k = \sin(kx)}$$

$$u(t,x) = \sum_{k=1}^{\infty} c_k e^{-k^4} \sin(kx)$$

$$u(0,x) = \sum_{k=1}^{\infty} c_k \sin(kx) = \sin x \overset{\text{koefs}}{\Longrightarrow} c_k = 0 \text{ za } k \neq 1$$
$$u(t,x) = e^{-t} \sin x$$

Milhom de sons

martela

1

$$u_{tt} - c^2 u_{xx} = F(t,x)$$
$$u_x(t,0) = t^2$$
$$u(t,L) = -t$$
$$u(0,x) = x^2 - L^2$$
$$u_t(0,x) = \sin^2\left(\frac{\pi x}{L}\right)$$

u_1, u_2 rešitvi

$v = u_1 - u_2$

$v \equiv 0$

$$v_{tt} - c^2 v_{xx} = u_{1tt} - u_{2tt} - c^2 u_{1xx} + c^2 u_{2xx} =$$
$$= F(t,x) - F(t,x) = 0$$

$$v_x(t,0) = t^2 - t^2 = 0$$
$$v(t,L) = -t + t = 0$$
$$v(0,x) = 0$$
$$v_t(0,x) = 0$$

$$\varphi(t) = \int_0^L \left(v_t^2 + c^2 v_x^2\right) dx$$

$$\varphi'(t) = \int_0^L \left(2 v_t v_{tt} + 2c^2 v_{xt} v_x\right) dx =$$

$$= 2\int_0^L v_t v_{tt}\, dx + 2c^2 \int v_{xt} v_x\, dx$$

$u = v_x \quad dv = v_{xt}$
$du = v_{xx} \quad v = v_t$

$$= 2\int_0^L v_t v_{tt}\, dx + 2c^2\left(v_x v_t - \int_0^L v_{xx} v_t\, dx\right) =$$

$$v_{tt} = c^2 v_{xx}$$

$$c^2 2\int_0^L v_t v_{xx} + 2c^2 v_x v_t\Big|_0^L - 2c^2\int_0^L v_{xx} v_t\, dx =$$

$$= 2c^2\left(v_x(t,L)\cdot v_t(t,L) - v_x(t,0)\, v_t(t,0)\right) =$$
$$= 0$$

$$\Rightarrow \varphi(t) = \varphi(0) = \int_0^L \left(v_t^2(0,x) + c^2 v_x^2(0,x)\right) dx =$$

$$= \int_0^L \left(0 + c^2 \cdot 0\right) dx = 0$$

$$\Rightarrow v_t^2 + v_x^2 = 0 \Rightarrow v_x = 0 \quad v_t = 0$$

$$\rightsquigarrow v = C(y) \wedge v = C(x) \Rightarrow v = C$$

konstanta

$\Rightarrow v = 0$ zaradi robnih pogojev

What is a solution to PDE

$$F(x, u, \nabla u, \dots) = 0$$

A solution would be a function u s.t.

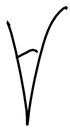
$$F(x, u, \nabla u, \dots) = 0$$

Since the equation involves derivatives of u up to some order $n \in \mathbb{N}$, we usually assume that the derivatives exist and are continuous

Def: A classical solution on an open set $\Omega \subseteq \mathbb{R}^2$ is $u \in C^n(\Omega)$ s.t. $F(x, u, \nabla u, \dots) = 0$

Standard questions:
- existence
- uniqunes
- maximal Ω

$\Bigg\}$ usualy related to the regularity of F



1. ura

7.5

$u, v \in \mathcal{C}^{1,2}((0,T)\times(0,L)) \cap C([0,T]\times[0,L])$

$u_t - u_{xx} \le v_t - v_{xx}$ na $(0,T)\times(0,L)$

$u \le v$ na paraboličnem robu

$\{0\}\times[0,L] \cup [0,T]\times\{0,L\}$

$\quad = \Gamma_T$

$\underline{u \le v \quad \text{na } [0,T]\times[0,L] = \Omega_T}$

Obravnavajmo $u_t - v_t - u_{xx} + v_{xx} =$

$= (u-v)_t - (u-v)_{xx} \le 0$

na robu: $u - v \le 0$
paraboličnem

$\Rightarrow \max\limits_{\Omega_{T-\varepsilon}} (u-v) = \max\limits_{\Gamma_{T-\varepsilon}}$

$\max\limits_{\Omega_T} (u-v) = \max\limits_{\partial\Omega_T} (u-v) \le 0$

$\Rightarrow \quad u \le v \quad$ na Ω_T

Harmonične funkcije nimajo izoliranih ničel $n=2$

$u_{xx} + u_{yy} = 0$ $\qquad$ $u(x_0, y_0) = 0$

$$u(x,y) = u(x+iy) = \frac{1}{2\pi} \int_{S_r} u(\alpha,\beta)\, dS_{\alpha,\beta}$$

Če je izolirana ničla, obstaja dovolj majhna okolica, ki nima ničel, torej u ne spremeni predznaka

$$\frac{1}{2\pi r} \int_0^1 \underbrace{u(\gamma(t))}_{\neq 0} \underbrace{|\dot{\gamma}(t)|}_{> 0} dt \neq 0$$

$$\Delta u(x) = 0 \Rightarrow \Delta u(Mx+b) = 0 \quad \forall b \in \mathbb{R}^n \text{ in}$$

$\forall M$ ortogonalne

$$\Delta(u(x)) = \sum_{i=1}^{n} u_{x_i x_i}(x)$$

$$\Delta(u(Mx+b)) = \sum_{i=1}^{n} \partial_{x_i}^2 u(Mx+b) =$$

$$= \sum_{i=1}^{n} \partial_{x_i}^2 \left(u([\sum m_{jk} x_k + b_j]_j) \right) =$$

$$= \sum \partial_{x_i} \left(\sum_j u_{x_j}([\dots]_j) \cdot m_{ji} \right) =$$

$$= \sum_i \sum_j \sum_l u_{x_j x_l}(Mx+b) \cdot \underbrace{m_{ji} \cdot m_{li}}_{\substack{=0 \; j \neq l \\ =1 \; j=l}} \overset{M \cdot M^T = I}{=}$$

$$= \sum_j \sum_l u_{x_j x_l}(Mx+b) I_{jl} =$$

$$= \sum_i u_{x_i x_i}(Mx+b) = \underset{0}{\underset{\shortparallel}{(\Delta u)}}(Mx+b) = 0$$

② $P(x,y) = \sum_{j=0}^{n} a_j x^j y^{n-j}$

$V_n = \mathrm{lin}\{ \cancel{P(x,y)}^{\text{harmonske}} ;\ a \in \mathbb{R}\}$ $\quad \dim V_n = ?$

$$\Delta P(x,y) = \partial \sum a_j \left(j x^{j-1} y^{n-j} + (n-j) x^j y^{n-j-1} \right) =$$

$$= \sum a_j \left(j(j-1) x^{j-2} y^{n-j} + (n-j) j x^{j-1} y^{n-j-1} + j(n-j) x^{j-1} y^{n-j-1} + (n-j)(n-j-1) x^j y^{n-j-2} \right) =$$

$$= \sum_{j=2}^{n} a_j x^{j-2} y^{n-j-2} \left(\underbrace{j(j-1)x^2 + 2j(n-j)xy + (n-j)(n-j-1)x^2}_{\neq 0} \right)$$

$V_0 \Rightarrow$ konst $\Rightarrow \dim V = 1$

$V_1 \Rightarrow a_0 x + a_1 y \Rightarrow \dim V = 2$

Pah, nekje sem se zmotila

Ponovimo

$$\Delta u = 0 \quad \text{na} \quad D$$
$$u = f \quad \text{na} \quad \partial D$$

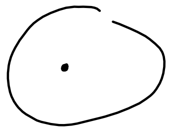
$$P_r(t) = \frac{1-r^2}{1-2r\cos(2\pi t)+r^2}$$

$$\Rightarrow u(r\cos(2\pi t), r\sin(2\pi t)) = (P_r * f)(t) = \int f(\tau) P_r(t-\tau)\,d\tau$$

(3) $U \subseteq \mathbb{R}^2$ $a \in U$

$\forall$ omejeno harm funk $u \in \mathcal{C}^2(U - \{a\})$

razširimo do harmonične na U



vzamemo majhen disk $D \subset U$

Označimo: $f = u|_{\partial D}$ razširimo f do harmonične

na U z u' $\Delta v = 0$ na D

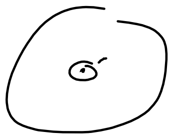
$u - v \equiv 0$

$(u-v)|_{\partial D} = 0$

$\Delta(u-v) = 0$ če je $u-v$ zvezna $\Rightarrow u \equiv v$

Rešitev na D $\Delta(\overset{w}{\widetilde{u-v}}) = 0$ ne

Recimo da v a ne moremo razširiti do zveznosti



izberemo α z

tej imamo 0

w na $D - K(a,r)$ je 0 za $\forall r$

w doseže maksimum na $K(a,r)$

ko pošljemo proti a mora biti

maksimum dosežen v a

$\log|z-a|$ $\log\left(\frac{|z-a|}{\varepsilon^2}\right)$

$w_\delta = w + \delta \cdot \log\left|\frac{z-a}{\varepsilon^2}\right|$ $\delta > 0$

ker je w omejena

$\max_{\partial K(a,r)} w_\delta \leq 0$ za dovolj majhen δ

$\Rightarrow w_\delta \leq 0$ na vseh D_r dovolj majhen δ

$w_\delta \leq 0$ na $D - \{a\}$

pošljemo $\delta \to 0 \Rightarrow w \leq 0$ na $D - \{a\}$

za $\delta < 0$ je $w_\delta \geq 0$

$\Rightarrow \delta \to 0$ $w \geq 0$

$\Rightarrow w = 0$ na $D - \{a\} \Rightarrow u$ lahko

razširimo na U

ⓤ

u zvezna do broba $u \geq 0$

$$\frac{1-|\rho|}{1+|\rho|} u(0,0) \leq u(\rho) \leq \frac{1+|\rho|}{1-|\rho|} u(0,0)$$

za $\forall \rho \in D$

razširimo do $\mathbb{D}$ z ρ

$|\rho| \to 1 \quad |\rho| e^{2\pi i t} \rightsquigarrow e^{2\pi i t}$

$$\rho : r e^{2\pi i t} \longmapsto \frac{r}{|\rho|} e^{2\pi i t}$$

$u = V \circ \rho = F \qquad u(0,0) = V(0,0)$

od 0 do ρ

$$\frac{1-|\rho|}{1+|\rho|} V(0,0) \leq V(e^{2\pi i t}) \leq \frac{1+|\rho|}{1-|\rho|} V(0,0)$$

$$V(re^{2\pi i t}) = P_r \circ u(t)\Big|_{|\rho|=1} = \int_0^1 u(\rho e^{2\pi i \tau}) P_r(t-\tau)\, d\tau =$$

$$= \int_0^1 u(\rho e^{2\pi i \tau}) \frac{1-|r^2|}{1-2|r|\cos(2\pi(t-\tau)) + |r|^2}\, d\tau$$

$$V(0,0) = \int_0^1 u(\rho e^{2\pi i \tau})\, d\tau$$

$$u(re^{2\pi i t}) = \int_0^1 u(e^{2\pi i \tau}) \frac{1-|\rho|^2}{1-2|\rho|\cdot\cos(2\pi(t-\tau))+|\rho|^2}\, d\tau$$

$$\frac{1-|\rho|^2}{1-2|\rho|\cos(2\pi(t-\tau))+|\rho|^2} \leq \frac{1-|\rho|^2}{1-2|\rho|+|\rho|^2} = \frac{\cancel{(1-|\rho|)}(1+|\rho|)}{(1-|\rho|)^2} = \frac{1+|\rho|}{1-|\rho|}$$

$$\frac{1-|\rho|^2}{1-2|\rho|\cos(2\pi(t-\tau))+|\rho|^2} \geq \frac{1-|\rho|^2}{1+|\rho|^2}$$

$$\int_0^1 u(e^{2\pi i t})\, d\tau \cdot C_1 = u(0,0)\cdot C_1$$

. . .   . . .

$\Longrightarrow$ odgovor

⑤ $\mathbb{H} = \mathbb{R} \times (0,\infty)$ $\quad u \in \mathcal{C}(\overline{H}) \cap C^2(\mathbb{H})$

$$\sup_{\overline{H}}(u) = \sup_{\partial H} u$$

Naj bo $\Omega(r) = \{ x, y \in \mathbb{H} ; \ x^2 + y^2 < r^2 \}$

Vemo: $u|_\Omega$ doseže maksimum na $\partial\Omega$

$v = u + \gamma \varphi$

Potrebujemo $\liminf \xrightarrow[r \to \infty]{} -\infty$

$$\varphi = -\log(r) = -\log(x^2 + (y+1)^2)$$

ista kot prej

$\Delta u = 0$ на $\mathbb{H}$

$u = f$ на $\partial \mathbb{H}$

$\varphi : \mathbb{H} \longrightarrow \mathbb{D}$

$z \longmapsto \frac{z-i}{z+i}$

$$v(re^{2\pi i t}) = \int_0^1 \tilde{f}(e^{2\pi i \tau}) P_r(t-\tau)\, d\tau = v(\varphi(z)) = u(z)$$

$$u(z) = v(\varphi(z)) =$$

$$P_r(t) = \frac{1-r^2}{1-2r\cos(2\pi t)+r^2}$$

$u(x,0) = 0$

u harmonične

$$v(x,y) = \begin{cases} u(x,y) & y \geq 0 \\ -u(x,-y) & y \leq 0 \end{cases}$$

$$\Delta v = \begin{cases} u_{xx} + u_{yy} & y > 0 \\ -u_{xx} - u_{yy} & y \leq 0 \end{cases} = 0$$

Bojč to ni vredu

Ali ima v LPV?

preveriti moramo samo na $(x,0)$ $y=0$

$$\frac{1}{2\pi r}\int_0^1 v(\underset{\overset{\|}{x_0}}{z_0} + re^{2\pi i t})\,dt = 0 \quad \text{za } \forall r \leq r_0 =$$

$$= \frac{1}{2\pi r}\left(\int_0^{\frac{1}{2}} u(z_0 + re^{2\pi i t})\,dt - \int_{\frac{1}{2}}^1 u(z_0 + re^{-2\pi i t})\,dt\right)$$

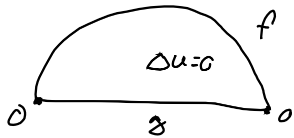
$$t = \tau + \tfrac{1}{2}$$

$$= \frac{1}{2\pi r}\left(\int_0^{\frac{1}{2}} \dots\,dt - \int_0^{\frac{1}{2}} u(z_0 + re^{-2\pi i \tau} \cdot e^{-\pi i})\,d\tau\right) =$$

$$= \frac{1}{2\pi r}\left(\int_0^{\frac{1}{2}} \dots\,dt - \int_0^{\frac{1}{2}} u(z_0 + re^{2\pi i \tau})\,d\tau\right) = 0 \quad \ddot\smile$$

$\Delta u = 0$ na B^+

$u = f$ na $\partial B_1(0) \cap$



ideja: razširimo pravilo na $\mathbb{B}$

∂D

$$v = \begin{cases} f & ; y > 0 \\ h & y < 0 \end{cases}$$

$$v(x,y) = f * P_r = \int_0^1 f(t-\tau) \frac{1-r^2}{1-2r\cos(2\pi i t) + r^2} d\tau$$

$$= \int_0^{\frac{1}{2}} f(\tau) \frac{1-r^2}{1-2r\cos(2\pi i(t-\tau)) + r^2} d\tau +$$

$$+ \int_{\frac{1}{2}}^1 h(\tau) \frac{1-r^2}{1-2r\cos(2\pi i t-\tau) + r^2} d\tau$$

vstavimo $y = 0$

$$\int_0^{\frac{1}{2}} f(\tau) \frac{1-r^2}{1-2r\cos\cos(2\pi i\tau) + r^2} d\tau +$$

$$+ \int_{\frac{1}{2}}^1 h(\tau) \frac{1-r^2}{1-2r\cos(2\pi\tau) + r^2} d\tau = g(r)$$

Rešitev za 

$\Delta v_2 = 0$ na D

$v_2 = \tilde{f}$ ∂D

Rešitev za

$\Delta v_1 = 0$

$v_1 = \tilde{g}$ na ∂H

Ideja rešimo za v_1 H in popravimo pogoje za v_2

$$v_1 = \frac{y}{\pi} \cdot \int_{-\infty}^{\infty} \frac{1}{(x-s)^2+y^2} \tilde{g}(s,0)\, ds$$ ... reprezentacijska formula

$$v_1 = \frac{y}{\pi} \int_{-1}^{1} \frac{1}{(x-s)^2+y^2} g(s,0)\, ds$$

$\Delta v_2 = 0$ na $\mathbb{H}$

$v_2 = \hat{f} - v_1$

$$v_2 = re^{i\vartheta} = (P_r * (\hat{f} - v_1))(e^{i\vartheta})$$

$w = v_1 + v_2$ reši problem. samo preveriti moramo

$v_2 = 0$ na $D \cap \{y=0\}$

$$v_2(x,0) = v_2(re^{i0}) = \frac{1}{2\pi} \int_0^{2\pi} (\hat{f} - v_1)(\cos t, \sin t) P_r(t)\, dt$$

$$P_r(t) \quad \frac{1-r^2}{1+r^2-2r\cos t}$$

$$\int_0^{2\pi} (\hat{f} - v_1)(\cos(t), \sin(t)) \cdot P_r(\tau)\, d\tau$$